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KCGM GROWTH PROJECT

DRAWING NO.
2000-F19-TLD-EL-10005

REV.: 0

DESIGN APPR.: *M Brown*

DATE: 31/08/23

MAIN CIRCUIT BREAKER

415V SUPPLY FROM MCC 19-SW-61
TIER 9F/3
REF. DWG. 19\E\1157-005

L1 L2 L3 N E

EARTH BAR

NEUTRAL BAR

DISTRIBUTION BOARD NO. 60 EMERGENCY

415V 3 PHASE+N 50Hz 100A FAULT LEVEL 18kA FOR 0.3 sec.
POSITIONED IN SUB STATION No.60 ON SWITCHBOARD 19-SW-61 TIER 9F/4

NOTE: THE FEEDER FOR THIS BOARD AUTOMATICALLY SWITCHES TO AN EMERGENCY SUPPLY IN THE EVENT OF A POWER OUTAGE. REFER TO DRAWINGS 19\E\1157-004 & 80\E\1132 FOR DETAILS.

SUB STATION No.60 LIGHTING	16A	R1	R2	20A	SUB STATION No.60 CYCLO CONVERTER ROOM LIGHTING
SUB STATION No.60 LIGHTING	16A	W1	W2	16A	SUB STATION No.60 CYCLO CONVERTER ROOM LIGHTING
SPARE		B1	B2	16A	FIRE ALARM
SUB STATION No.60 EMERGENCY LIGHTING	16A	R3	R4	16A	FIM SAG AREA SAFETY SHOWER
BLUE LIGHT SCATS	16A	W3	W4	16A	SPARE
SUB STATION No.60 EMERGENCY LIGHTING	16A	B3	B4		SPACE
KNELSON LIGHTING	16A	R5	R6	16A	TUNNEL LIGHTS
KNELSON LIGHTING	16A	W5	W6	16A	TUNNEL LIGHTS EAST
SAG MILL E-HOUSE LIGHTING AND SMALL POWER DISTRIBUTION BOARD F19-DB-72 19\E\1160-007	50A	B5	B6	16A	TUNNEL LIGHTS WEST
		R7	R8	16A	SUB STATION No.60 VESDA - 19FDP6003A
		W7	W8	16A	SUB STATION No.60 VESDA - 19FDP6003B
SUBSTATION LIGHTING (STH ENTRANCE)	16A	B7	B8	6A	PHOTO ELECTRIC CELL

A
63A
BYPASS

P.E.
SIGNAL

C2
C1
N

100A

EARTH LINK
NEUTRAL LINK
P
G/Y Bu W R

TO BATTERY BACKED-UP EMERGENCY LIGHT FITTINGS

PRIMERO HAVE IDENTIFIED THERE'S POSSIBILITY OF ADDING NEW LIGHTS ON THE EXISTING CIRCUIT 'AB5' WHICH FEEDS NEAR-BY LIGHTING CIRCUIT. ELECTRICAL CONTRACTOR TO CONFIRM THIS CIRCUIT IS ADEQUATE TO ADD NEW 4x FLOODLIGHTS & 6x60W WALKWAY LIGHTS

0

AREA LIGHTING 19CV03 BALL CHARGER	16A	R1	R2	16A	FLOOD LIGHTS NORTH SIDE
AREA LIGHTING	16A	W1	W2	16A	AREA LIGHTING CV53
TRANSFORMER AND SUB STATION No.60 WEST DOOR LIGHTS	16A	B1	B2	16A	AREA LIGHTING NTH SYD
AB5 LIGHTING CIRCUIT	16A	R3	R4	16A	AREA LIGHTING MTC LUBE
AREA LIGHTING CV23	16A	W3	W4	16A	AREA LIGHTING
AREA LIGHTING	16A	B3	B4	16A	FLOOD LIGHTS SOUTH SIDE
FLOOD LIGHTS SOUTH SIDE	16A	R5	R6	16A	AREA LIGHTING
AREA LIGHTING MTC TRASH	16A	W5	W6	16A	AREA LIGHTING
CV23 AREA LIGHTING	16A	B5	B6		SPACE
CV20 AREA LIGHTING	16A	R7	R8		SPACE
KNELSON AREA LIGHTING	16A	W7	W8	16A	SPARE
KNELSON AREA LIGHTING	16A	B7	B8	16A	SPARE

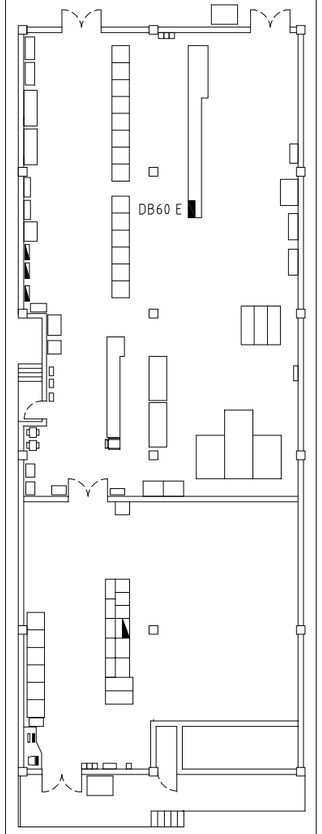
EMERGENCY LIGHT FITTINGS ARE FED FROM THIS DB.
THEY ARE THE "BATTERY BACKED-UP" TYPE OF FITTING.
THEY ARE FED WITH TWO FEEDS FROM THE SAME CB.
ONE FEED IS SWITCHED THROUGH C2 PE CELL CONTACTOR,
THE OTHER FEED IS DIRECT TO THE FITTING. THIS ALLOWS
FOR THE LIGHTS TO BE SWITCHED ON A DAYLIGHT SWITCH
WITH THE DIRECT FEED MAINTAINING THE BATTERY CHARGE.
IN THE EVENT OF A POWER OUTAGE THE BACK-UP BATTERY
WILL POWER THE LIGHT UNTIL POWER IS RESTORED.

LEGEND

- CIRCUIT BREAKER
- ISOLATOR
- CIRCUIT BREAKER RCD 30mA
- CONTACTOR
- CONTACTOR COIL

LOCATION

SUB STATION 60
19-DB-60 EMERGENCY



NOTES

TT - PROTEUS DWG No.
215033-DE-024

2000-F19-TLD-EL-10005_Rev0
Power schematic